

Health & Safety

Standard Earthworks



Reference

Company	Document Type	Department	N°	Version	Langue
BYTP	REF	H&S	2226	B	EN

Version	Date	Written by	Checked by	Approved	Reason for modification
A	01-11-2023	O.BRZEZINSKI	I.LELIEVRE	L.KNOLL	First issue
B	21-05-2024	A.JANSSENS	O.DANNAU	L.KNOLL	Modification of 1 basics Modification and merging of 2 Digging and shielding and embankments
C	19-12-2025	I.MUCCIGNATO	F.MARTIN	L.KNOLL	Modification of 2 digging- Safe access/exits every 15 m

The Standards presented in this document serve as a practical guide to define the means to be deployed for earthwork activities.

The elements of this document are complementing the elements presented in the following Standards:

- Standard Vehicles pedestrian interaction [BYTP-REF-H&S-2212]
- Standard Construction machinery and associated maintenance [BYTP-REF-H&S-2215]
- Standard Site Installations [BYTP-REF-H&S-2216]

Table of contents

1. Life-Saving rules vehicles – pedestrian interaction	4. Earthworks large mass (1/2)
2. Digging	5. Earthworks large mass (2/2)
2.1 Shielding	6. “Vertical” earthworks (wells , stations , etc.)
2.2 Embankments	7. Permit to dig
3. Construction sites	



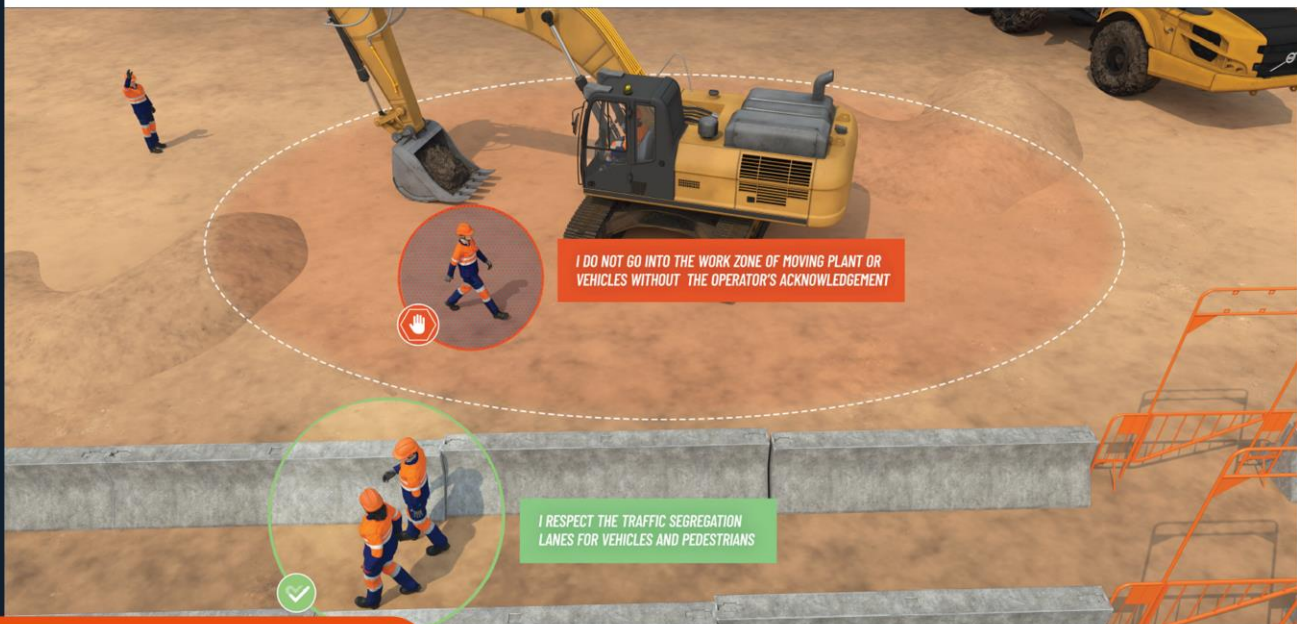
1. Life-Saving rules vehicles – pedestrian interaction



RISK COLLISION



- ✓ I respect the **traffic segregation lanes** for vehicles and pedestrians
- ✓ I do not go into the work zone of moving plant or vehicles without the **operator's acknowledgement**
- ✓ I do not enter an **exclusion zone**



If unable to adhere to these rules, I "STOP" and alert my hierarchy.

SAFETY
CULTURE

2. Digging

Before the beginning of the operation:

- Be in possession of the network layout plans indicating the location and type of underground pipes or cables in the work area
- Ensure the identification of networks on site (marking / pegging out of networks)
- Define the precautions required to secure the digging. For any vertical excavation of more than 1.3 m depth and width less than 2/3 of the depth = shielding (failing this, excavate in steps, justified embankment)
- Define a minimum distance between the circulation of vehicles/machines and the edge of the excavation following a geotechnical study and a risk analysis or failing that apply the following:
 - 1.5m for a machine with tracks
 - 2m for a wheeled vehicle
 - 2.5m for a wheeled and loaded vehicle

Depending on the geology of the soil, a calculation note by the technical department may be required.

During the operation:

- Respect all measures identified during the preparation phase (shielding, distance for the circulation, etc.)
- Provide safe access to excavations (trenches) every 15 metres.

Excavation or access not secured = access strictly forbidden

Excavation and access secured = access only to authorized persons

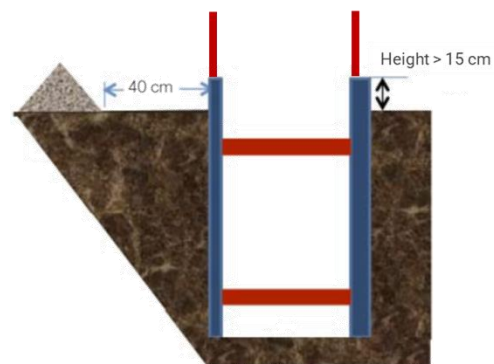
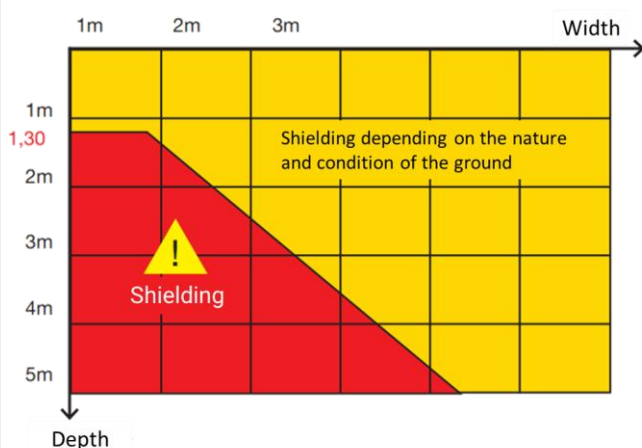
- Stop in case of unplanned event : unexpected presence of a network, hazardous waste, war device, etc.

Discovery of a war device = Immediate stop of activity / Emergency situation procedure activated

- After a period of rain or frost, the embankments, excavations or trenches are checked by a competent person (TWE).



2.1. Shielding



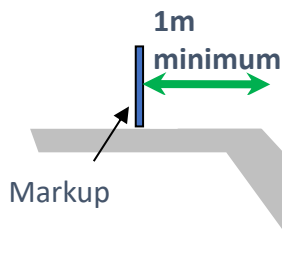
- Provide collective protection on shielding (if it is at less than 1m from pedestrian traffic)

2.2. Embankments

The angle of the slope must be done according to the recommendation of the organization which carried out the soil study and must be validated by the Technical Department (TWE). Ensure that there is no overload at the head of the slope which were not foreseen in the calculations (location of mobile cranes, positioning of storage, etc.).

The slope of an embankment does not exceed 1:1 = 45°.

Pedestrian traffic must be protected at the head of the embankment:

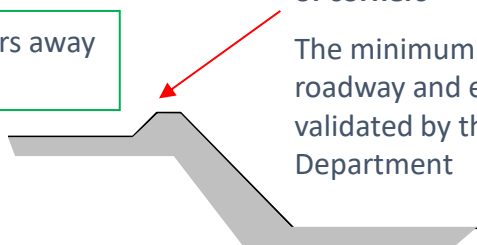


- Distance $\geq 1 \text{ m}$ = Marking of the embankment ridge (chain, etc.)
- Distance $< 1 \text{ m}$ = Compliant rigid collective protection (guardrail + plinth)



Vehicle circulation must be protected at the head of the embankment using a merlon / wheel guard or failing that by marking at the head of the embankment:

Storage/parking = at least at 2 meters away from the embankment



Compulsory wheel chaser outside of corners

The minimum distance between roadway and embankment is validated by the Technical Department

The minimum height of the wheel chaser must be greater than or equal to half of the biggest wheel used on the site.

In absence of merlon, storage forbidden at the head of the embankment

Specific measures for rain must be implemented:

- Digging gutters at the head and the bottom of the embankment
- Drainage

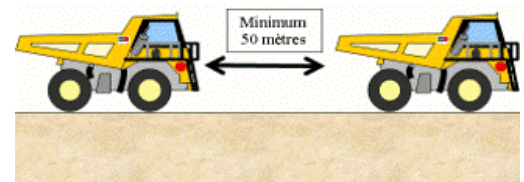
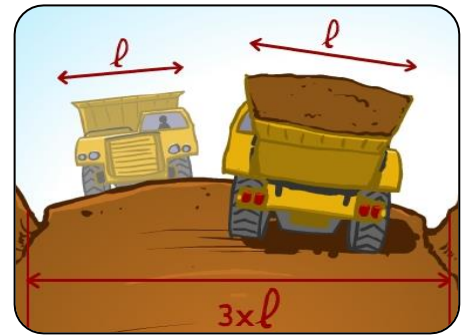
If necessary, and additionnaly to the above measures, ensure the covering of embankments (to prevent scouring due to runoff water) and any other measure required to allow the stability of the slope.



3. Construction sites

Traffic rules

- Distance between 2 machines = 50m
- Useful track width = 3 times the width of the machine
- Traffic man required when crossing a public road
- Consider potential coactivity
- Rotating beacon, reversing alarm and lights on
- Maximum speed to be defined according to the specificities of the sites
- If possible, separate the different traffic flows (dumpers, heavy goods vehicles, light vehicles, etc.)



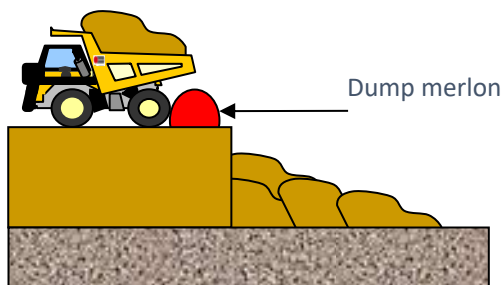
The taxiways must be

- Bodyworked taking into account climatic phenomena (water flow, rain, etc.), type of materials, geotextile, etc.
- Inspected and maintained regularly
- Watered if necessary (dust limitation). Be careful not to water the ramps!



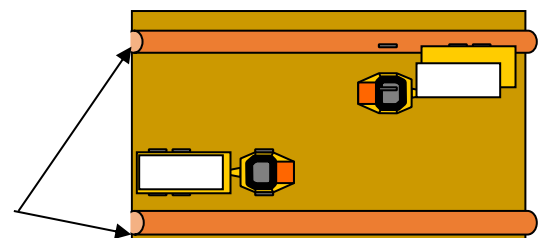
Use of geotextile

Any singular point (crossing of tracks, road crossing, aerial network, work workshop, narrowing requiring priority, etc.) must be signaled or protected.



Earth barrier intended to stop trucks at the edge of the embankment

Safety merlon



Wheel chase with a height at least equal to half the height of the largest wheel of construction machinery

4. Large mass earthworks – 1/2

The cutting

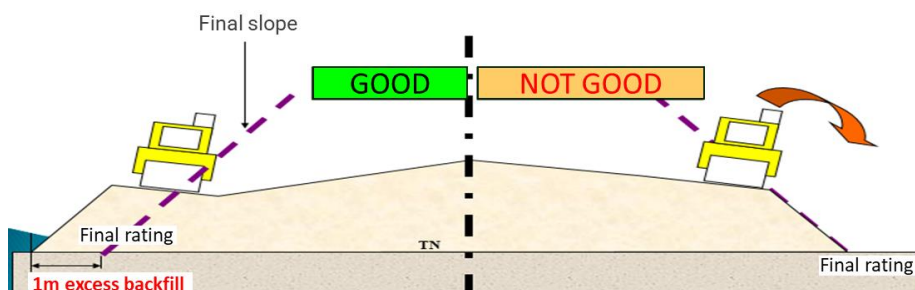
- Do not undercavate
- Ensure that the berms are wide and strong enough for equipment access

Scour protection



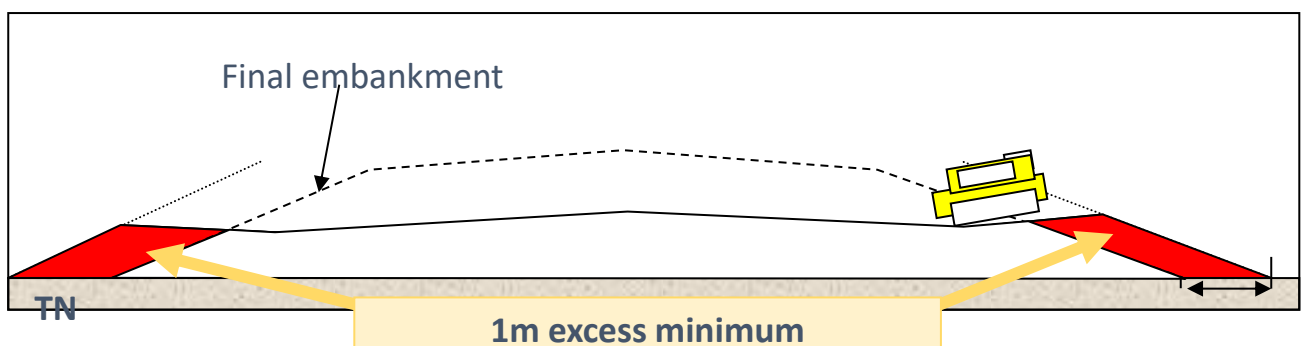
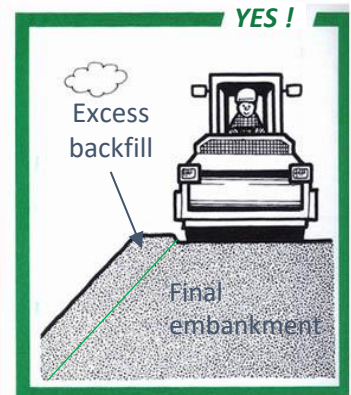
The fill

The use of excess fill allows safe compaction of the edge of the fill by reversing the slope to avoid tipping the compactor



For new embankments, the materials will be implemented using the so-called W method, thus allowing good compaction and working in complete safety (inward slope).

It is necessary to follow the technique of excess backfill (1m minimum) in order to avoid the compactors tipping over.

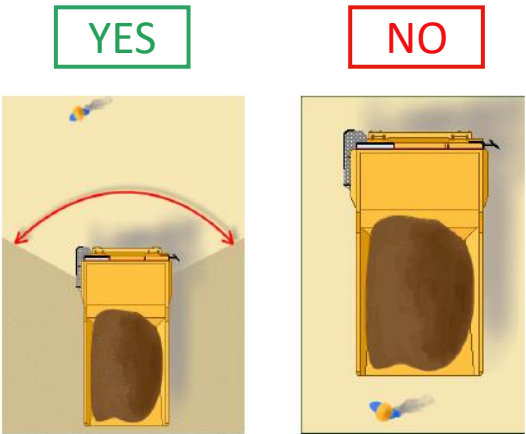
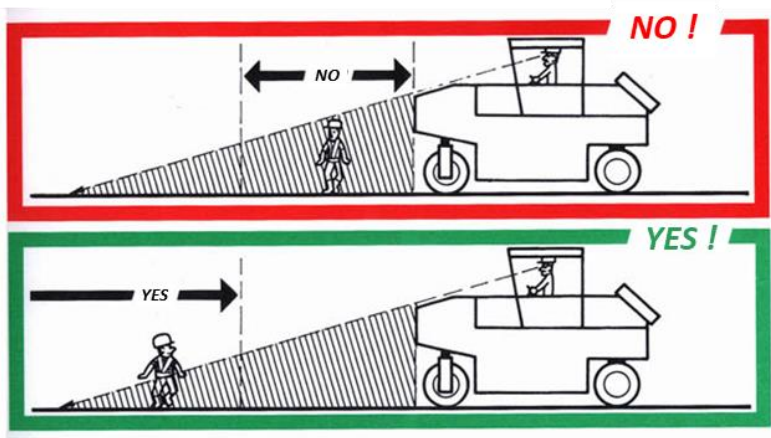


5. Large mass earthworks – 2/2

Machine guidance

Perform preferably the maneuvering of earth-moving equipment, without pedestrian in the vicinity.

- The traffic man must be trained and qualified
- Guidance is always done from the front to be seen by the driver, taking into account the blind spots of the machines
- Mandatory guidance if significant coactivity, cramped site, etc.
- Create a protective merlon to stop vehicles reversing



Conventional gestures

 Take command or watch out!	 Indicate a direction	 Indicate a distance behind the vehicle	 Move forward	 Stop the vehicle
 To move back	 Lift the skip	 Lower the skip	 Stop the skip	 End of command

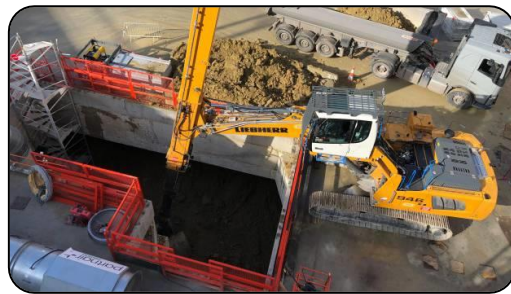
6. “Vertical” earthworks (wells, stations, etc.)

Use of machines

No human presence at the bottom during the earthworks phases:



- With a grab excavator
- With a crawler excavator cameleon
- Etc.



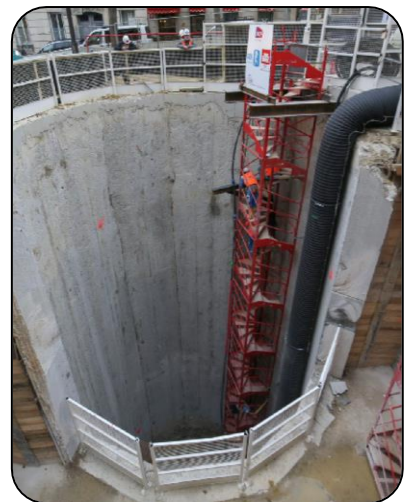
The cabins of earthmoving machines located at the bottom of the well are equipped with grille-type protection.

The Site Installation Plan must integrate

- The position of earthmoving equipment on the surface
- The buffer stock of excavated material
- The positioning of trucks responsible for removing debris (including a waiting area)
- The possible interference between cranes

Well equipment to be planned

- Secured access for personnel working at the bottom of the excavation
- Ventilation if applicable (depth, surface area, etc.)
- Downhole gas detection, based on specific risk analysis (polluted soils, exhaust gases, etc.)
- First aid equipment
- Protection of personnel at the bottom of the shaft during lifting operations



7. Permit to dig

In certain working environments, the complexity of the work and the regulation in place may require the establishment of a Permit to dig signed and issued by a designated and competent person.

A single permit-to-work may be issued for multiple identical operations covered by the same design, on the basis that each operation is appropriately checked prior to the operation commencing.

A permit to work is established before commencing any excavation work, if one of the following risks have been identified:

- Possibility of water ingress
- Presence of foundations close to the excavation or adjacent structures
- Loose rock or soil (including soil previously excavated)
- Presence of utilities in the ground (gas, electricity, water)
- Possibility of contaminated soil (hazardous substances or unexploded devices)
- Presence of simultaneous operations, and the possibility of cave-ins (considering the following potential events: sliding, toppling, bulging, heaving, boiling of the ground, and flooding)